Tutorial 8: Equivalence Class and Boundary Value Testing

Answer 1(a): Equivalence classes and boundary values

Humidity

	Invalid	Valid	Invalid
Equivalence classes	0 – 34	35 – 50	51 – 100

Valid equivalence class

Value on boundary values	35, 50
Value just below the boundary	34
Value just above the boundary	51

Invalid equivalence classes

Value on boundary values	0, 34
Value just below the boundary	-1
Value just above the boundary	35

Value on boundary values	51, 100
Value just below the boundary	50
Value just above the boundary	101

Answer 1(a): Equivalence classes and boundary values (cont’d)

Temperature

	Invalid	Valid	Invalid
Equivalence classes	-10 – 39	40 – 60	61 – 100

Valid equivalence class

Value on boundary values	40, 60
Value just below the boundary	39
Value just above the boundary	61

Invalid equivalence classes

Value on boundary values	-10, 39
Value just below the boundary	-11
Value just above the boundary	40

Value on boundary values	61, 100
Value just below the boundary	60
Value just above the boundary	101

Answer 1(b): Test cases

Valid Humidity Boundary Values (35, 50) + Valid Temperature Boundary Values (40, 60)

35 40	**Accept**
35 60	Accept
50 40	Accept
50 60	Accept

Answer 1(b): Test cases (cont'd)

Invalid Humidity (0, 34, 51, 100) + Valid Temperature (40, 60)

0 40	Reject
0 60	Reject
34 40	Reject
34 60	Reject
51 40	Reject
51 60	Reject
100 40	Reject
100 60	Reject

Answer 1(b): Test cases (cont'd)

Valid Humidity (35, 50) + Invalid Temperature (-10, 39, 61, 100)

35 -10	Reject
35 39	Reject
35 61	Reject
35 100	Reject
50 -10	Reject
50 39	Reject
50 61	Reject
50 100	Reject

Answer 1(b): Test cases (cont'd)

If assume the sensors can return invalid data (Humidity <= -1 or Humidity >= 101; Temperature <= -11 or Temperature >= 101), need to test the following cases:

-1 40	Exception: invalid data
-1 60	Exception: invalid data
101 40	Exception: invalid data
101 60	Exception: invalid data
35 -11	Exception: invalid data
35 101	Exception: invalid data
50 -11	Exception: invalid data
50 101	Exception: invalid data